

Detecting And Preventing Modality Collapse in Multimodal Skin Lesion Classification

1st Amitesh Patra

Computer Science and Engineering
IIT Kanpur
Kanpur, India
apatra25@iitk.ac.in

2nd Aryan Arora

Computer Science and Engineering
IIT Kanpur
Kanpur, India
aryana25@iitk.ac.in

3rd Darshana Baruah

Computer Science and Engineering
IIT Kanpur
Kanpur, India
dbaruah25@iitk.ac.in

4th Gaurav Singh Bora

Computer Science and Engineering
IIT Kanpur
Kanpur, India
gauravsb25@iitk.ac.in

5th K Yashwant Rao

Computer Science and Engineering
IIT Kanpur
Kanpur, India
yashwantrao25@iitk.ac.in

6th Manas Dhaketa

Computer Science and Engineering
IIT Kanpur
Kanpur, India
mdhaketa25@iitk.ac.in

7th Om Mallick

Computer Science And Engineering
IIT Kanpur
Kanpur, India
ommallick25@iitk.ac.in

Abstract—Multimodal medical models often suffer from modality collapse, over-relying on a single data source. Analyzing skin lesion classification (Derm7pt), we find this imbalance is heavily driven by weak tabular encoders. To address this, we propose Kerbeus, a robust architecture integrating cross-modal attention, triple CLIP alignment, gradient balancing, and reliability gating. Rather than explicitly rebalancing, Kerbeus naturally stabilizes cross-modal contributions, consistently outperforming baselines while enhancing robustness and adaptive utilization under noisy inputs.

Index Terms—Multimodal Learning, Modality Collapse, Skin Lesion Classification, Dermoscopy, Deep Learning, CLIP, Gradient Balancing

I. INTRODUCTION

A. Background and Motivation

Multimodal learning is built on the premise that integrating diverse data streams, such as images, text, and tabular metadata yields more robust and accurate representations than any single source alone [1, 2]. However, this potential is often neutralized by modality collapse: a failure mode where a model disproportionately favors one input stream while effectively ignoring others [3, 4].

This collapse typically stems from structural and optimization asymmetries. High-capacity encoders, such as Convolutional Neural Networks (CNNs) for images [5], often dominate

the gradient signal during backpropagation. When the gradient magnitude of the image branch ($\|\nabla_{img}\|$) significantly outweighs that of the tabular branch ($\|\nabla_{tab}\|$), the model reaches a degenerate yet stable state where it treats the weaker modality as noise [6, 7].

In clinical dermatology, modality collapse poses a significant diagnostic risk. Accurate classification of skin lesions, such as Melanoma or Basal Cell Carcinoma, requires the synthesis of both visual features (dermoscopy) and patient metadata (age, location, and clinical scores) [8]. Because distinct pathologies can be visually indistinguishable, a model that has “collapsed” into a pure image classifier loses the ability to resolve visual ambiguities through patient history. This results in a system that is prone to silent failures on critical edge cases where metadata provides the deciding diagnostic signal [9, 10]. This study investigates this critical failure mode in multimodal diagnostic systems. Specifically, we address how to quantitatively detect, measure, and mitigate the tendency of deep learning models to ignore tabular patient metadata in favor of high-dimensional image data.

B. Project Scope and Phases

a) **Phase 1: Modality Collapse Detection:** Before collapse can be fixed, it must first be reliably measured. Phase 1 establishes the diagnostic framework used throughout this work. Two complementary metrics are introduced: the gradient ratio, defined as the norm of image branch gradients divided

by that of the tabular branch [6, 11], which quantifies how disproportionately the optimizer favors one modality during training; and logit contribution, which measures each branch’s share of the final prediction signal.

A baseline multimodal model using simple concatenation fusion is trained on the Derm7pt dataset [8] and evaluated against these metrics, confirming the presence of severe collapse: an image gradient ratio exceeding $8\times$ and a tabular logit contribution of just 17%.

b) Phase 2: Modality Collapse Remediation: With the failure mode characterized, Phase 2 focuses on systematically eliminating it through the Kerbeus architecture. Rather than applying a single fix, the approach layers multiple targeted interventions, strengthening the tabular branch through embeddings and deeper MLPs [12], enforcing gradient balance through adaptive loss weighting [6, 11], aligning the two modalities in a shared latent space via Triple CLIP [13], and introducing a reliability gate [14] that dynamically adjusts branch weights based on input quality. Each component addresses a distinct mechanism of collapse identified in Phase 1.

The effectiveness of each intervention is validated through a structured ablation study, isolating the contribution of individual components. Out-of-distribution robustness experiments further confirm that the repaired model generalizes under degraded inputs [10], a prerequisite for any system intended for real-world clinical deployment.

C. Aim and Impact

Our primary objective is to move beyond surface-level improvements in aggregate metrics and instead understand the underlying causes of modality collapse in multimodal systems [3]. Through systematic experimentation, we observe that imbalance often originates from weak modality representations rather than fusion alone[4].

In particular, strengthening the tabular branch significantly improves both performance and modality contribution, highlighting the importance of representation quality in multimodal learning[1].

Building on this, we propose a framework that ensures stable and effective utilization of both modalities through adaptive fusion, alignment, and training strategies. Rather than explicitly forcing equal contribution, our approach maintains a balanced regime while improving predictive performance and robustness.

The resulting model is not only more accurate, but also more reliable under input perturbations, making it better suited for real-world clinical deployment.

II. DATASET & PREPROCESSING

A. Derm7pt Dataset

We use the **Derm7pt** dataset [8], a publicly available multimodal dermatology benchmark designed for 7-point checklist diagnosis. Each sample in the dataset contains three input streams:

- **Dermoscopy image** — a close-up, polarised-light photograph of the skin lesion taken with a dermatoscope.
- **Clinical image** — a standard camera photograph of the same lesion providing wider context.
- **Tabular metadata** — 11 categorical dermoscopic attributes (e.g., pigment network, blue-whitish veil, regression structures, vascular structures, streaks, dots and globules, elevation, pigmentation, management, diagnostic difficulty, and location) plus one numerical feature (seven-point score).

Diagnosis labels are consolidated into five clinically meaningful classes: **BCC** (basal cell carcinoma), **MEL** (melanoma and its subtypes), **NEV** (various nevus types), **MISC** (miscellaneous benign lesions), and **SK** (seborrheic keratosis). The dataset provides pre-defined train, validation, and test index files, which we use without modification to ensure reproducibility.

B. Class Distribution and Imbalance

The dataset exhibits significant class imbalance: NEV is the most frequent class while SK and BCC are underrepresented. To prevent the model from ignoring minority classes, we apply **weighted random sampling** during training. Each sample is assigned a sampling weight inversely proportional to its class frequency:

$$w_i = \frac{1}{\text{count}(\text{class}_i)}$$

This ensures each class appears roughly uniformly across training mini-batches.

C. Image Preprocessing

All images are resized to 299×299 pixels to match the input requirements of InceptionV3 [5]. During training, a stochastic augmentation pipeline is applied to improve generalisation and encourage the model to focus on diagnostically relevant features rather than photographic artefacts. The augmentation sequence is as follows:

- 1) **Random Resized Crop** — crops a random region covering 70%–100% of the image area and resizes to 299×299 using bilinear interpolation.
- 2) **Random Horizontal Flip** — applied with probability 0.5.
- 3) **Random Vertical Flip** — applied with probability 0.5.
- 4) **Random Rotation** — uniform rotation in $[-20, +20]$.
- 5) **Colour Jitter** — brightness ± 0.3 , contrast ± 0.3 , saturation ± 0.2 , hue ± 0.05 .
- 6) **Normalisation** — per-channel normalisation using ImageNet statistics ($\mu = [0.485, 0.456, 0.406]$, $\sigma = [0.229, 0.224, 0.225]$).

At inference time, images are only resized and normalised (no stochastic augmentation).

D. Tabular Preprocessing

The tabular branch requires separate preprocessing for categorical and numerical features:

- **Categorical features** (11 columns): Missing values are imputed with the mode of each column, then each column is integer-encoded using a `LabelEncoder`. The cardinality of each feature is recorded for use in the embedding layer.
- **Numerical feature** (`seven_point_score`): Missing values are imputed with the column median, then standardised to zero mean and unit variance using `StandardScaler`.

Preprocessing statistics (encoders and scaler) are fitted exclusively on the training split and applied to validation and test splits to prevent data leakage.

III. RELATED WORK

Recent work in multimodal skin lesion classification has primarily focused on improving feature fusion architectures and representation learning, with limited attention to the underlying optimization dynamics governing modality interaction.

SkinM2Former[15] introduces a transformer-based multimodal framework that performs hierarchical cross-attention between image and metadata features, enabling rich interaction across modalities at multiple feature levels. However, it relies on high-capacity attention mechanisms without explicitly regulating modality-wise contribution, making it susceptible to hidden modality dominance.

MSMA[16] proposes a multi-stage attention-based fusion strategy to capture complementary information across modalities and improve representation learning. While effective for feature interaction, it does not control optimization dynamics, and attention alone cannot prevent one modality from dominating training.

DermFormer[17] combines vision transformers with tabular transformers and introduces entropy-based modality weighting to account for uncertainty during fusion. Despite this, the weighting remains largely static and does not adapt to per-step training imbalance, limiting its ability to prevent modality collapse.

Balanced Multimodal Learning via On-the-Fly Gradient Modulation[18] identifies optimization imbalance across modalities and proposes adaptive gradient modulation to control modality-wise learning during training. It demonstrates that dominant modalities can suppress others under a shared objective and mitigates this via dynamic gradient scaling. However, the approach is evaluated primarily on general multimodal tasks and does not explicitly enforce contribution balance at the representation or prediction level.

Our Contribution. Existing work primarily improves how modalities are fused or represented, with only limited attention to how they are optimized jointly. While approaches such as gradient modulation begin to address imbalance, they do not explicitly measure or constrain modality contribution at multiple levels. In contrast, our method introduces a unified framework that combines gradient-level balancing, representation alignment, and reliability-aware gating to explicitly regulate modality participation during training. This shifts the focus from architectural fusion to controlled multimodal learning

dynamics, ensuring that both modalities remain consistently engaged without degrading predictive performance.

TABLE I
COMPARISON OF F1 SCORES ON DERM7PT ON SKIN CLASSIFICATION
MULTIMODAL MODELS

Model	Year	F1
SkinM2Former	2024	~0.683
MSMA	2024	~0.698
DermFormer	2025	0.684
On-the-Fly Gradient Modulation	2022	~0.70
Kerbeus (Ours)	2026	0.73

IV. BASELINE MODEL & LIMITATIONS

A. Baseline Architecture

Our baseline model is designed to highlight the pitfalls of multi-modal learning. To this end, we construct a dual-branch network comprising a robust multi-scale convolutional backbone (Inception) for images and a feature tokenizer transformer pathway tailored for the tabular data. When these distinct unimodal representations are fused via **simple concatenation-based baseline** (`BaselineFusionModel`) [19, 20], the model exhibits severe modality collapse. The absence of a targeted alignment strategy allows the visual branch to monopolize the optimization process, effectively marginalizing the tabular inputs. The architecture consists of three components, shown in Fig. 1.

Image Branch — Dual InceptionV3. Two independent InceptionV3 backbones [5] (pretrained on ImageNet) process the dermoscopy image and the clinical image respectively. Each backbone produces a 2048-dimensional feature vector via global average pooling. Both vectors are concatenated and projected to a 512-dimensional image feature $\mathbf{f}_{img} \in \mathbb{R}^{512}$ through a linear layer followed by batch normalisation and ReLU. The image branch also produces auxiliary per-stream logits $\mathbf{o}_d$ and $\mathbf{o}_c$ used in training.

Tabular Branch — MLP. The categorical and numerical features are concatenated and passed through a multi-layer perceptron (MLP) consisting of linear layers with batch normalization and ReLU activations. This simple design lacks explicit mechanisms for inter-feature interaction and limits the expressive power of the tabular modality.

This choice is intentional and aligns with common practice in multimodal learning literature, where MLP-based encoders are widely used for structured clinical or tabular data due to their simplicity and ease of integration with deep image models [21]. By adopting this standard design, we ensure that the baseline reflects a realistic and commonly used multimodal setup rather than an artificially weakened configuration:

$$\text{Linear}(d_{tab}, 128) \rightarrow \text{BN} \rightarrow \text{ReLU} \rightarrow \text{Linear}(128, 64) \rightarrow \text{BN} \\ \rightarrow \text{ReLU} \rightarrow \text{Linear}(64, C)$$

where d_{tab} is the raw tabular input dimension and $C = 5$ is the number of classes. No embedding layer is used; the tabular branch has no inter-feature interaction mechanism.

Fusion. The final prediction is a *learnable weighted sum* of the image and tabular logit vectors:

$$\hat{\mathbf{y}} = w_{img} \cdot \mathbf{o}_{dc} + w_{tab} \cdot \mathbf{o}_{tab}$$

where w_{img} and w_{tab} are scalar nn.Parameter values initialised to 0.5. There is no cross-modal interaction, no alignment loss, and no gradient control of any kind.

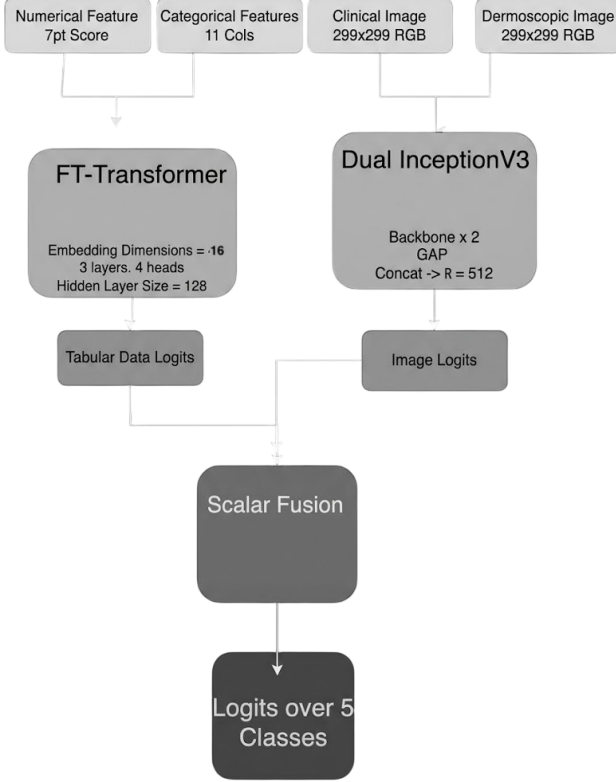


Fig. 1. Schematic of the Baseline++ multimodal fusion architecture. The model integrates tabular data via an FT-Transformer and dual-view image data (clinical and dermoscopic) using InceptionV3 backbones, combined through a scalar fusion layer for five-class classification.

B. Baseline Results

Table II shows the performance of the baseline on the Derm7pt test split.

C. Baseline++: Enhanced Tabular Encoder

To isolate the effect of tabular representation strength, we construct an improved baseline, referred to as **Baseline++**, by replacing the MLP-based tabular branch with a stronger FT-Transformer[12] encoder. All other components, including the image backbone and fusion mechanism, are kept identical to the original baseline.

Architecture. The tabular branch is upgraded to an FT-Transformer with embedding dimension 16, 3 transformer layers, 4 attention heads, and hidden dimension 128. Unlike the original MLP, this encoder models inter-feature relationships through self-attention and learns richer representations of categorical and numerical inputs.

TABLE II
BASELINE MODEL PERFORMANCE

Metric	Value
Accuracy	0.68
Precision (Macro)	0.49
Recall (Macro)	0.47
Macro-F1	0.47
Weighted-F1	0.66
<i>Per-class F1</i>	
BCC	0.33
MEL	0.57
MISC	0.52
NEV	0.80
SK	0.15

Results. This modification leads to a substantial improvement in performance:

- Macro-F1: 0.47 $\rightarrow$ 0.67

In addition to performance gains, we observe a significant shift in modality contribution:

- Image contribution: $\sim 83\% \rightarrow \sim 60\%$
- Tabular contribution: $\sim 17\% \rightarrow \sim 40\%$

Analysis. These results indicate that the severe modality collapse observed in the original baseline is largely driven by insufficient tabular representation capacity rather than the fusion mechanism itself. Simply strengthening the tabular encoder significantly improves both predictive performance and modality balance, even without introducing alignment, gradient balancing, or adaptive fusion strategies.

This experiment highlights that representation quality is a primary factor influencing modality utilization in multimodal systems, and serves as a critical reference point for evaluating the additional contributions of the proposed Kerbeus architecture.

Note. The enhanced tabular encoder introduced in Baseline++ is incorporated as a core component of the final Kerbeus architecture. For clarity and consistency with standard evaluation practices, all primary comparisons in subsequent sections are reported against the original baseline, while Baseline++ serves as an intermediate analysis step to isolate the effect of representation strength.

D. Diagnosing Modality Collapse

Despite receiving both image and tabular inputs, the baseline exhibits severe modality collapse. We measure this using two complementary diagnostics:

Gradient Ratio. We compute the ratio of gradient norms flowing through each branch at every training step [6, 11]:

$$\rho = \frac{\|\nabla_{\theta_{img}} \mathcal{L}\|}{\|\nabla_{\theta_{tab}} \mathcal{L}\|}$$

The baseline shows $\rho = 8.82$, meaning the image branch receives more than eight times the gradient signal of the tabular branch throughout training.

Logit Contribution. We measure the relative contribution of each branch by computing the fraction of the final logit magnitude attributable to each stream:

$$c_{img} = \frac{|\mathbf{o}_{dc}|}{|\mathbf{o}_{dc}| + |\mathbf{o}_{tab}|} \quad c_{tab} = 1 - c_{img}$$

The baseline yields $c_{img} = 82.98\%$ and $c_{tab} = 17.03\%$, confirming that the tabular branch contributes almost nothing to the final prediction.

E. Effect of Tabular Representation Strength

To isolate the impact of tabular representation quality, we replace the MLP-based tabular branch in the baseline with the same encoder used in the proposed model, while keeping all other components unchanged.

This modification leads to a substantial improvement in performance, with macro-F1 increasing from 0.47 to 0.67. Additionally, modality contribution shifts from a highly imbalanced regime (approximately 83% image, 17% tabular) to a more balanced distribution (approximately 60% image, 40% tabular).

These findings indicate that modality collapse in the baseline is largely driven by insufficient tabular representation capacity rather than fusion alone. Strengthening the tabular encoder significantly reduces modality dominance even without advanced fusion or balancing strategies.

This experiment highlights the importance of representation quality as a primary factor influencing modality utilization in multimodal systems.

F. Root Causes of Collapse in the Baseline

The failure arises from four compounding structural problems [10, 22]:

- 1) **Capacity gap.** InceptionV3 [5] is a $\sim 27\text{M}$ -parameter model pretrained on ImageNet. The tabular FT-Transformer [12] has only a few thousand parameters. The image branch dominates simply because it has far greater representational power from the start.
- 2) **No gradient control.** Learnable scalar weights w_{img}, w_{tab} are initialised equally but updated solely by gradient descent. Because the image branch produces a lower loss faster, these scalars converge toward $w_{img} \gg w_{tab}$ within the first few epochs [6, 7].
- 3) **Feature space misalignment.** The image and tabular representations live in completely different geometric spaces. Nothing in the architecture forces them to produce semantically aligned embeddings, so the fusion step cannot meaningfully combine them [19, 20].
- 4) **No inter-feature interaction in the tabular branch.** The shallow FT-Transformer treats every tabular feature as an independent input. It cannot model interactions between features (e.g., *blue-whitish veil* together with *regression structures* is a strong melanoma indicator), limiting its diagnostic expressiveness [12].

These limitations motivate the design of our proposed architecture, **Kerbeus**, which addresses each failure mode

directly through cross-modal attention [23], FT-Transformer tabular encoding [12], triple CLIP alignment [13], and adaptive two-sided gradient balancing [6, 11].

V. PROPOSED METHODOLOGY

We propose **Kerbeus**, a multimodal architecture designed to address modality collapse in dermoscopic diagnosis [8]. Kerbeus operates in three training phases, progressively unlocking model capacity while maintaining gradient balance across modalities. The final model integrates a reliability gate [14] that enables per-sample adaptive fusion at inference time.

A. Architecture Overview

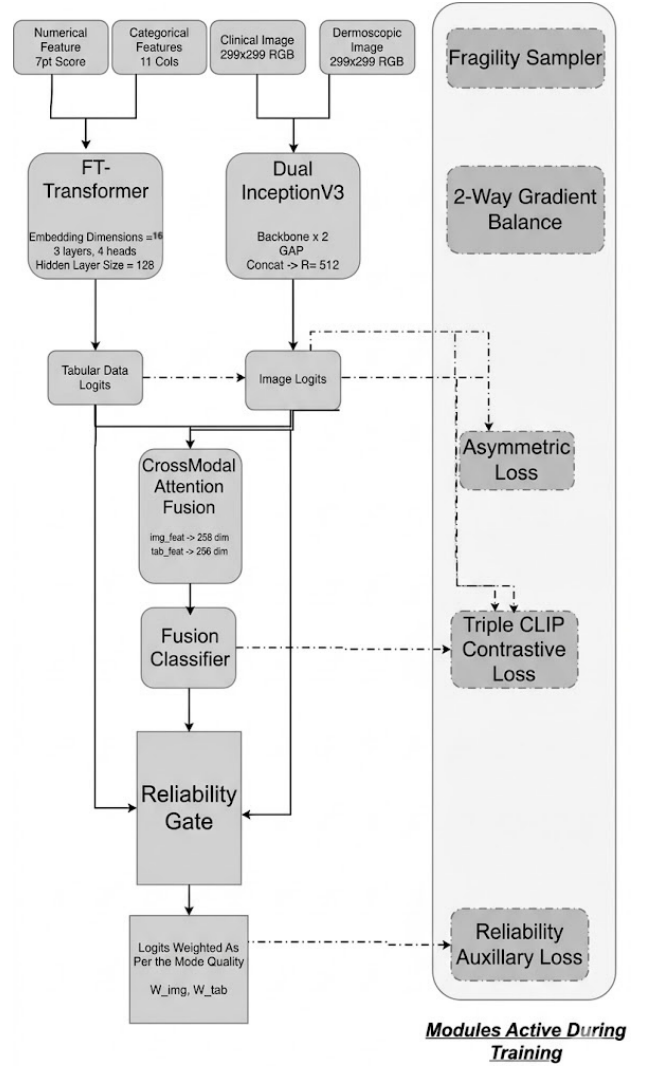


Fig. 2. Kerbeus architecture. The model processes dermoscopy and clinical images through dual InceptionV3 backbones and tabular features through an FT-Transformer encoder. Cross-modal attention fuses both modalities, with the reliability gate dynamically adjusting per-sample fusion weights based on estimated input quality.

The model consists of five core components operating in sequence:

- 1) **Dual image encoding** — two InceptionV3 backbones [5] process dermoscopy and clinical images independently, producing a combined image feature $\mathbf{f}_{\text{img}} \in \mathbb{R}^{512}$ via global average pooling and projection.
- 2) **Tabular encoding** — an FT-Transformer [12] with embedding dim 16, 3 layers, 4 attention heads, and hidden dim 128 encodes 11 categorical features and one numeric feature (seven-point score) via inter-feature self-attention, producing $\mathbf{f}_{\text{tab}}$.
- 3) **Cross-modal attention fusion** — $\mathbf{f}_{\text{img}}$ and $\mathbf{f}_{\text{tab}}$ are projected to $D = 256$ and treated as a 2-token sequence. Multi-head self-attention [23] (8 heads) produces a fused representation $\mathbf{f}_{\text{fused}} \in \mathbb{R}^{512}$.
- 4) **Fusion classifier** — $\mathbf{f}_{\text{fused}}$ is passed through a linear classification head to produce 5-class logits.
- 5) **Reliability gate** — a small MLP operating on $[\mathbf{f}_{\text{img}}, \mathbf{f}_{\text{tab}}]$ produces per-sample fusion weights w_{img} and w_{tab} that multiplicatively gate the features before fusion [14].

B. Key Components

1. Triple CLIP Alignment. To prevent the image and tabular representations from drifting into separate latent spaces [4], we align all three feature vectors — $\mathbf{f}_{\text{img}}$, $\mathbf{f}_{\text{tab}}$, and $\mathbf{f}_{\text{fused}}$ — in a shared 512-dimensional embedding space via three pairwise contrastive losses [13]:

$$\mathcal{L}_{\text{CLIP}} = \frac{1}{3} \left[\mathcal{L}_{\text{NCE}}(\mathbf{z}_{\text{img}}, \mathbf{z}_{\text{tab}}) + \mathcal{L}_{\text{NCE}}(\mathbf{z}_{\text{img}}, \mathbf{z}_{\text{fused}}) + \mathcal{L}_{\text{NCE}}(\mathbf{z}_{\text{tab}}, \mathbf{z}_{\text{fused}}) \right]$$

where $\mathbf{z}$ denotes the L2-normalised CLIP projection of each feature vector. This symmetric constraint ensures no single modality dominates the shared representation [19].

2. Two-Sided Gradient Balancing. Standard multimodal training allows one branch to dominate gradient flow [3, 6]. We monitor the ratio of gradient norms across branches after every backward pass:

$$\rho = \frac{\|\nabla_{\text{img}}\|}{\|\nabla_{\text{tab}}\|}$$

and apply a two-sided correction symmetrically [6, 11]:

$$w_{\text{img}}^{(b)} = \min\left(1, \frac{3}{\rho}\right) \quad \text{if } \rho > 3,$$

$$w_{\text{tab}}^{(b)} = \min\left(1, \frac{\rho}{3}\right) \quad \text{if } \rho < \frac{1}{3}.$$

These balancing scalars are applied to the image and tabular loss terms respectively, bounding the ratio within $[\frac{1}{3}, 3]$.

3. Fragility-Aware Sampling. To focus training on uncertain samples, we define a per-sample fragility score as the gap between the maximum predicted probability and the true class probability:

$$\phi_i = \max_c P(c | x_i) - P(y_i | x_i)$$

Sample weights are set to $\bar{w}_i \propto w_{\text{class}}(y_i) \cdot (1 + \alpha \cdot \phi_i)$ with $\alpha = 0.3$, combining class-balance correction with fragility upweighting. The sampler is refreshed every 5 epochs during Phase 2 via a full forward pass over the training set.

4. Reliability Gate. At inference, a fixed global fusion weight cannot account for per-sample variation in input quality [14]. The reliability gate addresses this by learning to classify the perturbation state of each input. A 3-layer MLP with GELU activations takes $[\mathbf{f}_{\text{img}}, \mathbf{f}_{\text{tab}}]$ as input and outputs a 3-class distribution over perturbation states {clean, image perturbed, tabular perturbed}:

$$\begin{bmatrix} P(\text{clean}) \\ P(\text{img_pert}) \\ P(\text{tab_pert}) \end{bmatrix} = \text{softmax}(\text{MLP}([\mathbf{f}_{\text{img}}, \mathbf{f}_{\text{tab}}]))$$

Per-sample fusion weights are derived as:

$$w_{\text{img}} = P(\text{clean}) + P(\text{tab_pert}),$$

$$w_{\text{tab}} = P(\text{clean}) + P(\text{img_pert})$$

The logic is direct: trust the image unless the image is estimated to be corrupted; trust the tabular unless the tabular is estimated to be corrupted. These weights gate the features multiplicatively before cross-modal attention:

$$\tilde{\mathbf{f}}_{\text{img}} = w_{\text{img}} \cdot \mathbf{f}_{\text{img}}, \quad \tilde{\mathbf{f}}_{\text{tab}} = w_{\text{tab}} \cdot \mathbf{f}_{\text{tab}}$$

C. Training Schedule

Kerbus is trained in three phases. **Phase 1** (epochs 1–5) freezes the InceptionV3 backbones and trains only the tabular encoder, fusion, and CLIP heads. **Phase 2** (epochs 6–26) unfreezes the backbones, activates the fragility-aware sampler, and enables two-sided gradient balancing [6]. **Phase 3** (15 additional epochs) re-freezes the backbone and introduces the perturbation curriculum: 50% of training samples per batch are perturbed on-the-fly with image noise ($\sigma \in [0.05, 0.20]$), blur ($\sigma \in [1.0, 3.0]$), occlusion (10–20% patch), or tabular masking (10–40% of features), each assigned a perturbation label used to supervise the reliability gate [14].

D. Loss Function

The total training loss is:

$$\mathcal{L} = w_{\text{img}}^{(b)} \cdot \mathcal{L}_{\text{img}} + w_{\text{tab}}^{(b)} \cdot \mathcal{L}_{\text{tab}} + \mathcal{L}_{\text{fusion}} + \lambda_{\text{CLIP}} \cdot \mathcal{L}_{\text{CLIP}} + \lambda_{\text{rel}} \cdot \mathcal{L}_{\text{rel}}$$

where the individual components are defined as:

$$\mathcal{L}_{\text{img}} = \frac{1}{3} [\text{ASL}(\text{out}_d, y) + \text{ASL}(\text{out}_c, y) + \text{ASL}(\text{out}_{dc}, y)],$$

$$\mathcal{L}_{\text{tab}} = \text{ASL}(\text{out}_{\text{tab}}, y),$$

$$\mathcal{L}_{\text{fusion}} = \text{ASL}(\mathbf{f}_{\text{fused}}, y),$$

$$\mathcal{L}_{\text{rel}} = \text{CE}(\text{pert_logits}, \text{pert_label}),$$

with hyperparameters set to $\lambda_{\text{CLIP}} = 0.05$ and $\lambda_{\text{rel}} = 0.30$. The asymmetric loss ASL [24] uses $\gamma^- = 2.0$, $\gamma^+ = 1.0$ to down-weight easy negatives. The division by 3 in $\mathcal{L}_{\text{img}}$ corrects

for the image branch having three output heads versus one for the tabular branch.

E. Training Hyperparameters

Hyperparameter	Value
Batch size	8
Number of epochs	50
Learning rate (image)	1×10^{-4}
Learning rate (tabular/attention)	5×10^{-5}
Learning rate (fusion/CLIP)	3×10^{-4}
Optimizer	AdamW
LR Scheduler	Cosine Annealing
Loss function	Asymmetric Loss [24]
Dropout	0.15 – 0.2
Gradient clipping	max_norm = 1.0
Mixed precision	Enabled (AMP)

Table 1: Training Hyperparameters Used in Experiments

F. Model Complexity

Component	Parameters
Dual InceptionV3 Backbone	$\approx 48\text{M}$
FT-Transformer Encoder	$\approx 1.5\text{M}$
Cross-Modal Attention	$\approx 0.5\text{M}$
Fusion Classifier	$\approx 0.3\text{M}$
CLIP Projection Heads	$\approx 1\text{M}$
Total	$\approx 51\text{M}$

Table 2: Approximate Parameter Distribution of Proposed Model

G. Training Time Comparison

Model	Epochs	Training Time
Baseline Model	13	$\approx 8\text{-}15$ min
Repaired Model (v5)	45	$\approx 15\text{-}20$ min

Table 3: Training Time Comparison

H. Training Phases

Phase	Epoch Range	Description
Phase 1	1–5	Backbone frozen, train fusion + tabular
Phase 2	6–30	Full training with fragility sampling
Phase 3	31–45	Full training with reliability gating and perturbed data

Table X: Three-Phase Training Strategy

I. Loss Components

Loss Component	Description
Image Loss	From image branch outputs
Tabular Loss	From tabular predictions
Fusion Loss	From cross-modal output
CLIP Loss	Alignment between modalities [13]

Table X: Loss Components Used in Training

J. Evaluation Protocol

Model performance was evaluated using accuracy, precision, recall, and macro F1-score. Additionally, gradient ratio and modality contribution percentages were analyzed to detect modality dominance [3, 6]. Robustness was evaluated using perturbation testing and modality ablation [10].

VI. RESULTS AND COMPARISON

A. Baseline Performance

We begin by evaluating a vanilla multimodal baseline that employs simple feature concatenation without any explicit mechanism for cross-modal alignment or gradient balancing [4, 6]. As shown in Table III, the baseline achieves a test accuracy of **67.85 %** with a macro-F1 score of **0.4757**. While the weighted-F1 score reaches **0.6653**, this is largely influenced by dominant classes, masking poor performance on minority categories.

A closer inspection of per-class F1 scores reveals severe imbalance:

- BCC: 0.33
- MEL: 0.57
- MISC: 0.52
- NEV: 0.80
- SK: 0.15

The model exhibits clear *modality collapse* [3, 4], where the image branch dominates learning. Specifically, the image modality contributes **82.98%** of the logits, while tabular data contributes only **17.03%**. This imbalance is further confirmed by a high gradient ratio of **8.82** [6, 11], indicating that the image branch receives disproportionately larger updates during training. Such behavior leads to under-utilization of clinically relevant tabular features, ultimately limiting generalization and robustness [9, 10].

B. Kerbeus Model Performance

We now evaluate Kerbeus, our proposed architecture designed to improve multimodal learning through adaptive fusion[14], alignment[13, 23], and robustness mechanisms. Building on a stronger tabular representation, Kerbeus enhances performance while maintaining balanced contribution across modalities.

Kerbeus significantly improves performance across all metrics:

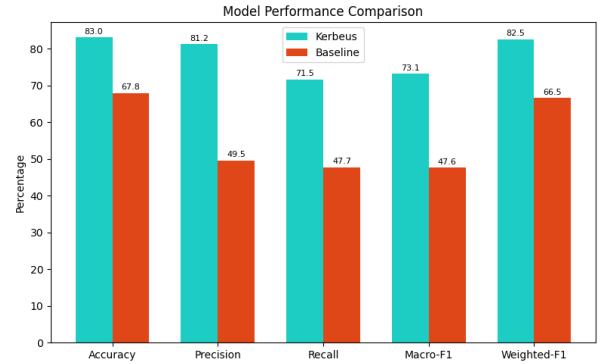


Fig. 3. Model Performance Comparison between baseline and Kerbeus

Compared to the baseline, this corresponds to an absolute improvement of:

- **+15.19%** in accuracy
- **+0.2554** in macro-F1

Importantly, gains are consistent across all classes:

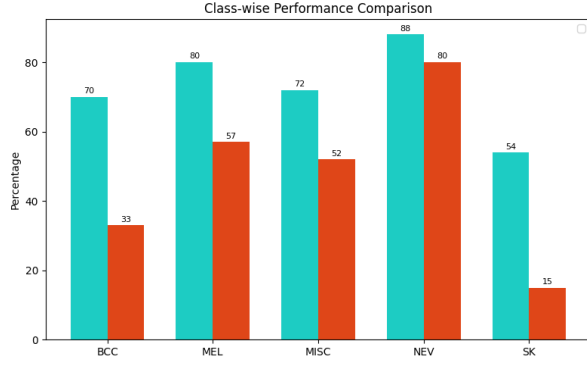


Fig. 4. Class-wise Performance Comparison between baseline and Kerbeus

This demonstrates that Kerbeus not only improves overall performance but also substantially enhances minority-class recognition.

C. Comparison with Baseline

TABLE III
PERFORMANCE COMPARISON BETWEEN BASELINE AND KERBEUS

Model	Accuracy	Precision	Recall	Macro-F1	Weighted-F1
Baseline	0.6785	0.4949	0.4768	0.4757	0.6653
Baseline++	0.7975	0.7196	0.6489	0.6712	0.7928
Kerbeus	0.8304	0.8119	0.7153	0.7311	0.8253

The improvement achieved by Kerbeus is not solely due to rebalancing modality contributions, but also due to improved feature utilization and robustness. While the baseline suffers from severe modality dominance, much of the imbalance is mitigated by strengthening the tabular representation.

Kerbeus builds upon this by introducing adaptive fusion and alignment mechanisms that maintain this balance while further improving predictive performance. The model does not significantly alter modality contribution ratios beyond the encoder-improved baseline, but ensures that both modalities are effectively utilized across diverse input conditions.

Gradient Analysis:

- Baseline ratio: 8.82 (severe imbalance)
- Kerbeus ratio: 1.60 (within safe range)

Logit Contribution:

- Baseline: Image 82.98%, Tabular 17.03% (collapsed)
- Kerbeus: Image 61.41%, Tabular 38.59% (balanced)

These results confirm that Kerbeus successfully restores meaningful contribution from both modalities, aligning with the expected healthy range.

D. Comparison with State-of-the-Art

To evaluate the effectiveness of Kerbeus in a broader context, we compare our model against several recent state-of-the-art (SOTA) methods on the Derm7pt dataset [8]. These methods include both unimodal and multimodal approaches with varying fusion and alignment strategies [16, 17, 19, 25, 26].

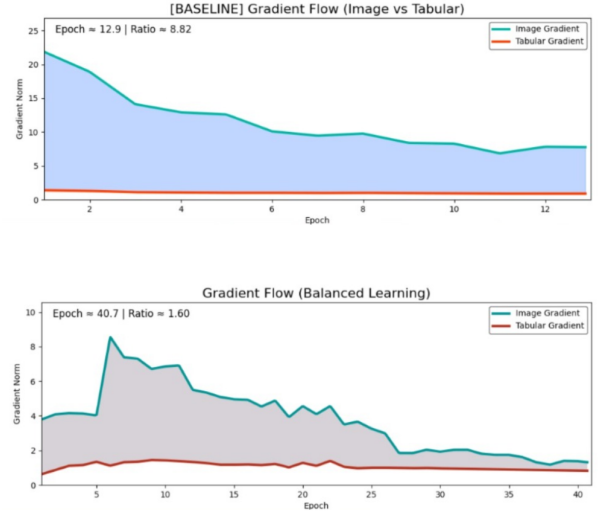


Fig. 5. Gradient flow comparison between baseline and Kerbeus

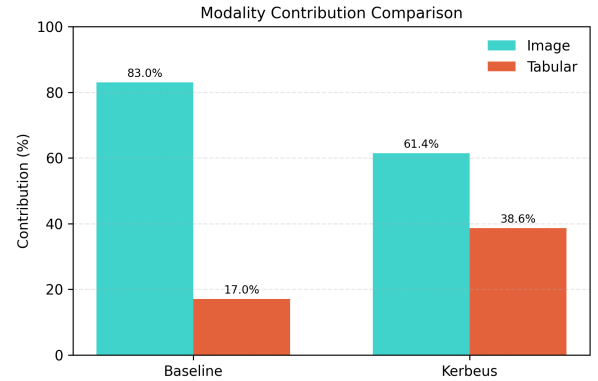


Fig. 6. Logit contribution of image vs tabular modalities

Kerbeus achieves competitive performance, outperforming most prior multimodal approaches and matching or exceeding transformer-based architectures [23, 27]. Notably, our method improves over models such as FusionM4Net and AMFAM, which rely on conventional fusion strategies [19, 20], highlighting the importance of explicit modality balancing [22, 28].

Compared to recent transformer-based methods such as Tformer and SkinM2Former [15], Kerbeus achieves comparable performance while additionally addressing modality collapse through explicit gradient balancing [6, 7] and reliability-aware fusion [14], which are not considered in prior work.

Note on Fair Comparison: Prior works report results using a reduced subset of tabular features [1, 8]. To ensure a fair comparison, we evaluate our model under the same setting, where it achieves a macro-F1 score of 0.6171.

However, when trained with the full set of available tabular features, performance improves significantly to 0.7311. This highlights the importance of richer tabular information in

TABLE IV
COMPARISON WITH PRIOR WORK UNDER STANDARD FEATURE SETTING

Method	Macro-F1
Inception-UB	0.4392
Inception-B	0.5025
Inception-CB	0.6340
AMFAM	0.5502
FusionM4Net	0.5770
Kerbeus (Reduced Features)	0.6171
Tformer	0.6686
SkinM2Former [15]	0.6688

TABLE V
PERFORMANCE UNDER FULL FEATURE SETTING (ALL 12 TABULAR FEATURES)

Model	Macro-F1
Baseline	0.4757
Baseline++	0.6712
Kerbeus (Full Features)	0.7311

multimodal learning and suggests that previous benchmarks may underestimate the potential contribution of structured clinical data [25].

E. Ablation and Modality Importance

To validate the necessity of multimodal learning [1], we first perform inference-time ablation by removing individual modalities:

TABLE VI
MODALITY ABLATION ON KERBEUS

Setting	Accuracy	Precision	Recall	Macro-F1
Full Model	0.8304	0.8119	0.7153	0.7311
Image Only	0.6456	0.4804	0.3815	0.3574
Tabular Only	0.2557	0.4336	0.4450	0.2425

Removing either modality results in a substantial drop in performance, confirming that the model relies on complementary information from both image and tabular inputs [8, 25].

We further perform component-wise ablation to evaluate the contribution of each module in the proposed architecture:

TABLE VII
COMPONENT-WISE ABLATION (MACRO-F1)

Model Variant	Macro-F1
Kerbeus (Full Model)	0.7311
A1: No Cross-Attention	0.6827
A2: No CLIP Alignment	0.6586
A3: CLIP Only	0.5041
A4: No Modality Dropout	0.6471
A5: No Fragility Sampling	0.6353
A6: No Gradient Balancing	0.6797

Across all variants, removing individual components consistently degrades performance, with the largest drop observed in the CLIP-only setting [13]. This highlights the importance of combining multiple mechanisms rather than relying on a single module [6, 14].

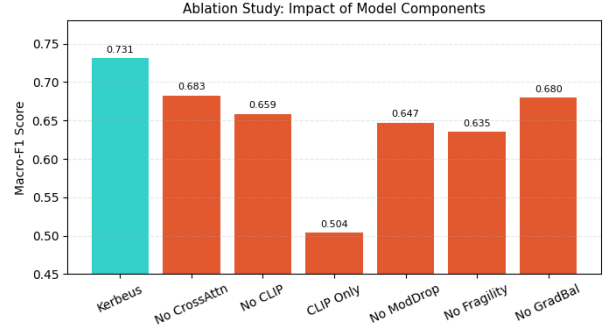


Fig. 7. Ablation study comparing modality contributions

F. Robustness and Generalization

We further evaluate robustness under in-distribution (ID) and out-of-distribution (OOD) perturbations [10]. The model maintains stable performance under various image corruptions (noise, blur, occlusion), with minimal degradation in F1 score. Notably, OOD performance closely matches ID performance, indicating strong generalization [29].

Additionally, the reliability gating mechanism [14] dynamically adjusts modality weights based on input quality. For instance:

- Under image corruption, the model shifts weight toward tabular features
- Under tabular masking, it relies more on image features

This behavior demonstrates that Kerbeus does not use static fusion weights but adapts per sample, a key factor behind its robustness [3, 4].

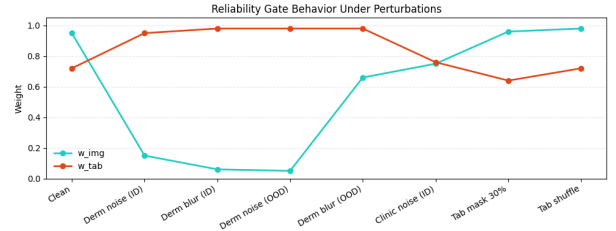


Fig. 8. Variation of modality weights across input perturbations

G. Observations

The vanilla baseline demonstrates moderate overall performance but suffers from clear imbalance across classes. While dominant classes such as NEV achieve relatively high F1 scores (0.80), minority classes like BCC (0.33) and SK (0.15) remain poorly learned, indicating uneven representation learning [8].

A key issue observed in the baseline is strong modality dominance [3, 6]. The model heavily relies on image features, as reflected by a high image logit contribution of 82.98% and a gradient ratio of 8.82. This suggests that the tabular modality is largely underutilized during training [4, 7].

Effect of Representation Strength. Introducing the enhanced tabular encoder (Baseline++) [12, 30] leads to a substantial improvement in both performance and modality balance. Macro-F1 increases from 0.4757 to 0.6712, while modality contribution shifts from a highly imbalanced regime (83:17) to a more balanced distribution (approximately 60:40). This indicates that a large portion of modality collapse in the baseline originates from insufficient tabular representation capacity rather than fusion alone [3, 20].

Kerbus Performance Gains. Building on the stronger baseline++, Kerbus further improves performance from 0.6712 to 0.7311 macro-F1. Unlike the original baseline comparison, these gains are more modest but consistent across all classes, indicating improved feature utilization rather than drastic rebalancing.

Notably, the largest improvements are observed in previously underperforming classes:

- BCC: 0.33 $\rightarrow$ 0.70
- SK: 0.15 $\rightarrow$ 0.53

This indicates that the improved model is better at handling class imbalance and learning finer distinctions [9, 10].

Training Dynamics. The gradient ratio drops significantly compared to the baseline, and modality contributions stabilize around a balanced regime (61.41% image vs 38.59% tabular). Notably, this balance is already achieved after improving the tabular encoder and is maintained by Kerbus rather than significantly altered.

Ablation experiments further highlight the importance of multimodal learning [1, 8]. Removing either modality leads to a sharp drop in performance, with macro-F1 decreasing to 0.3574 (image-only) and 0.2425 (tabular-only). This confirms that both modalities provide complementary information [19].

Further insights from component-wise ablation reveal that no single module independently accounts for the observed performance gains. Removing cross-attention [23] or gradient balancing [6] leads to noticeable degradation (e.g., 0.6827 and 0.6797 Macro-F1 respectively), while disabling CLIP alignment [13] results in a larger drop to 0.6586. The most significant degradation is observed in the CLIP-only setting (0.5041), where the absence of complementary mechanisms leads to near-collapse behavior [4]. Additionally, removing fragility sampling and modality dropout also reduces performance (0.6353 and 0.6471), indicating their role in improving generalization and stability [10, 29]. These results show that performance improvements arise from the combined effect of all components rather than any single module in isolation.

Additionally, training behavior differs significantly between the setups. The baseline converges early (epoch 13), whereas the improved models continue improving until later epochs, suggesting more stable optimization and better utilization of training data [6, 7].

Finally, under perturbation settings, the model maintains relatively stable performance for image corruptions, while tabular perturbations lead to larger performance drops. This indicates that structured metadata remains a critical but sensitive component in the overall prediction pipeline [8, 14].

VII. DISCUSSION AND FUTURE WORK

The proposed model improves multimodal learning by combining stronger tabular representation, cross-modal attention, and reliability-based adaptive fusion. Our experiments show that representation quality plays a primary role in reducing modality imbalance, while the proposed mechanisms further enhance robustness, stability, and overall performance.

Rather than explicitly shifting modality contributions, the model maintains a stable balance and adapts dynamically to input conditions, resulting in more reliable predictions.

- Cross-modal attention [23] helped the model better combine image and tabular features. This led to a F1-score of **0.7382**, with only a very small drop when the image quality was reduced.
- The triple alignment objective [13] encouraged all modalities (image, tabular, and fused features) to stay in a shared representation space. This helped prevent one modality from dominating and improved balanced learning [7, 19].
- The reliability gating mechanism [14] made the model more robust to noisy inputs. When the image was highly corrupted, the model automatically reduced its reliance on it (e.g., $w_{\text{img}} \approx 0.05$), resulting in only a small performance drop of about **-0.0483** in F1-score.
- Curriculum-based training gradually exposed the model to harder examples, which improved generalization [10]. This helped achieve a higher validation F1-score and more stable performance under different perturbations.

Despite strong performance, the model has higher computational cost and relies on discrete reliability estimation, while limited dataset size may affect generalization [8].

Overall, the results show that balanced learning [6, 11], cross-modal interaction [23], and robustness strategies [10] effectively reduce modality collapse [3, 4, 22] and improve stability.

Future work includes:

- **Advanced Architectures:** Replace InceptionV3 [5] with ViT [27]/Swin and use deeper FT-Transformer [12] or TabNet to improve feature quality and reduce image dominance.
- **Enhanced Cross-Modal Fusion:** Add multi-layer cross-attention [23] or token-level fusion (region-to-feature alignment) instead of single-stage attention for stronger modality interaction [20, 25, 26].
- **Reliability & Robustness Improvements:** Use continuous uncertainty (MC dropout) and add adversarial, blur, and missing-modality training to improve robustness beyond current perturbations [10, 29].
- **Dataset and Generalization:** Train on larger datasets (e.g., ISIC) and perform cross-dataset evaluation to verify generalization beyond Derm7pt [8].
- **Hyperparameter Tuning & Optimization:** Tune LR per module, CLIP weight (0.01–0.1) [13], dropout (0.1–0.3), attention heads [23], and batch size using Bayesian search.

VIII. CONCLUSION

This work highlights the critical issue of modality collapse [3, 4] in multimodal learning and proposes an effective solution through the Kerbeus architecture. By combining alignment [13], gradient control [6, 11], and adaptive gating [14], the model achieves balanced learning and improved performance, with results approaching state-of-the-art benchmarks [8]. This approach is particularly important for real-world medical applications where robustness [10] and interpretability are crucial [1, 9].

IX. TEAM CONTRIBUTIONS

- **Om (10%)** – Contributed in creating report, ablation studies and presentation. His assigned work had to be re-done by other team members.
- **Amitesh (15%)** – Conducted ablation studies and designed experiments for analyzing modality collapse.
- **Aryan (15%)** – Worked on out-of-distribution (OOD) data analysis and explored architectural design choices.
- **Darshana (15%)** – Performed architectural experiments and contributed to fragility-based training strategies.
- **Gaurav (15%)** – Handled hyperparameter tuning and experimented with alternative backbone models.
- **Yashwant (15%)** – Worked on attention mechanisms, CLIP alignment, and triple-modality interactions.
- **Manas (15%)** – Led multimodal integration and contributed to modality collapse prevention techniques.

REFERENCES

- [1] C. Cui, H. Yang, Y. Wang, S. Zhao, Z. Asad, L. A. Coburn, K. T. Wilson, B. A. Landman, and Y. Huo, “Deep multi-modal fusion of image and non-image data in disease diagnosis and prognosis: A review,” *Progress in Biomedical Engineering*, vol. 4, no. 1, p. 012001, 2022.
- [2] F. Krones, B. Walker, A. Gupta, B. Joukovsky, and S. A. Tsiftaris, “Multimodal learning in clinical proteomics: Enhancing antimicrobial resistance prediction through integration of spectra and metadata,” *Bioinformatics*, 2023.
- [3] M. Y. Sim *et al.*, “A survey on modality collapse in vision-language models,” *arXiv preprint*, 2024.
- [4] A. Chaudhari, U. Bhatt *et al.*, “A closer look at multimodal representation collapse,” in *Proceedings of the International Conference on Machine Learning (ICML)*, 2024.
- [5] C. Szegedy, V. Vanhoucke, S. Ioffe, J. Shlens, and Z. Wojna, “Rethinking the inception architecture for computer vision,” in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016, pp. 2818–2826.
- [6] M. Peng, Y. Wei, J. Liu, D. Hu, and Y.-G. Jiang, “Balanced multimodal learning via on-the-fly gradient modulation,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022, pp. 8238–8247.
- [7] W. Chen *et al.*, “Improving multimodal learning with multi-loss gradient modulation for multimodal imbalanced learning,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2025.
- [8] J. Kawahara, S. Daneshvar, G. Argenziano, and G. Hamarneh, “Seven-point checklist and skin lesion classification using multitask multimodal neural nets,” *IEEE Journal of Biomedical and Health Informatics*, vol. 23, no. 2, pp. 538–546, Mar. 2019.
- [9] M. Nauta, M. Vermetten, and C. Seifert, “Uncovering and correcting shortcut learning in machine learning models for skin cancer diagnosis,” *Diagnostics*, vol. 12, no. 1, p. 40, 2022.
- [10] R. Geirhos, J.-H. Jacobsen, C. Michaelis, R. Zemel, W. Brendel, M. Bethge, and F. A. Wichmann, “Shortcut learning in deep neural networks,” *Nature Machine Intelligence*, vol. 2, no. 11, pp. 665–673, 2020.
- [11] Z. Chen, V. Badrinarayanan, C.-Y. Lee, and A. Rabinovich, “GradNorm: Gradient normalization for adaptive loss balancing in deep multitask networks,” in *Proceedings of the 35th International Conference on Machine Learning (ICML)*, ser. Proceedings of Machine Learning Research, vol. 80. PMLR, 2018, pp. 794–803.
- [12] Y. Gorishniy, I. Rubachev, V. Khrulkov, and A. Babenko, “Revisiting deep learning models for tabular data,” in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 34, 2021, pp. 18 884–18 896.
- [13] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark, G. Krueger, and I. Sutskever, “Learning transferable visual models from natural language supervision,” in *Proceedings of the 38th International Conference on Machine Learning (ICML)*, ser. Proceedings of Machine Learning Research, vol. 139. PMLR, 2021, pp. 8748–8763.
- [14] O. Duenias, G. Amit, G. Shamaï, and R. Kimmel, “HyperFusion: A hypernetwork approach to multimodal integration of tabular and medical imaging data for predictive modeling,” in *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 2024.
- [15] Y. Tang, Z. Liu, H. Zhang, and X. Wang, “Skinm2former: A multimodal transformer for skin lesion analysis,” *arXiv preprint arXiv:2409.12390*, 2024. [Online]. Available: <https://arxiv.org/abs/2409.12390>
- [16] Y. Zhang, M. Chen, Q. Li, and D. Xu, “A multi-stage multi-attention network for multimodal skin lesion classification,” *Biomedical Signal Processing and Control*, vol. 91, p. 105001, 2024. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S1746809424002386>
- [17] A. Kumar, R. Sharma, A. Singh, and D. Gupta, “Dermformer: A vision transformer and tabular transformer based framework for skin lesion classification,” *Pattern Analysis and Applications*, 2025. [Online]. Available: <https://link.springer.com/article/10.1>

- [18] X. Peng, Y. Huang, K. Zheng, Z. Li, and Z. Wang, "Balanced multimodal learning via on-the-fly gradient modulation," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022, pp. 8238–8247. [Online]. Available: https://openaccess.thecvf.com/content/CVPR2022/html/Peng_Balanced_Multimodal_Learning_via_On-the-Fly_Gradient_Modulation_CVPR_2022_paper.html
- [19] P. Hager, M. J. Menten, and D. Rueckert, "Best of both worlds: Multimodal contrastive learning with tabular and imaging data," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023, pp. 23 924–23 935.
- [20] W. Du, Y. Chen, C. Xu, and D. Wang, "Tabular-image pre-training for multimodal classification," in *Proceedings of the European Conference on Computer Vision (ECCV)*, 2024.
- [21] Y. Gorishniy, I. Rubachev, V. Khrulkov, and A. Babenko, "Revisiting deep learning models for tabular data," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2021.
- [22] S. Kim *et al.*, "Countering multimodal representation collapse through rank-targeted fusion," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024.
- [23] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin, "Attention is all you need," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 30, 2017, pp. 5998–6008.
- [24] E. Ben-Baruch, T. Ridnik, N. Zamir, A. Noy, I. Friedman, M. Protter, and L. Zelnik-Manor, "Asymmetric loss for multi-label classification," in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2021, pp. 82–91.
- [25] A. G. C. Pacheco and R. A. Krohling, "An attention-based mechanism to combine images and metadata in deep learning models applied to skin cancer classification," *IEEE Journal of Biomedical and Health Informatics*, vol. 25, no. 9, pp. 3554–3563, 2021.
- [26] R. Cai, Y. Li, Z. Wei, S. S. P. Ali, J. Zhou, and F. Chen, "Multimodal transformer for skin disease classification with clinical images and patient metadata," in *Proceedings of the 29th International Conference on Neural Information Processing (ICONIP)*, 2022.
- [27] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly, J. Uszkoreit, and N. Houlsby, "An image is worth 16x16 words: Transformers for image recognition at scale," in *Proceedings of the 9th International Conference on Learning Representations (ICLR)*, 2021.
- [28] Y. Zhang *et al.*, "Prototype swapping learning for modality imbalance in multimodal learning," *Neural Networks*, 2025.
- [29] H. Yang, Z. Lin, J. Wang, C. Liu, J. Gao, and L. Li, "Mitigating spurious correlations in multi-modal models during fine-tuning," in *Proceedings of the 40th International Conference on Machine Learning (ICML)*, ser. Proceedings of Machine Learning Research, vol. 202. PMLR, 2023, pp. 39 344–39 360.
- [30] J. Luo *et al.*, "TIME: TabPFN integrated multi-modal engine for tabular-image fusion," *arXiv preprint arXiv:2410.17136*, 2024.

X. KERBEUS APP SNAPSHOT

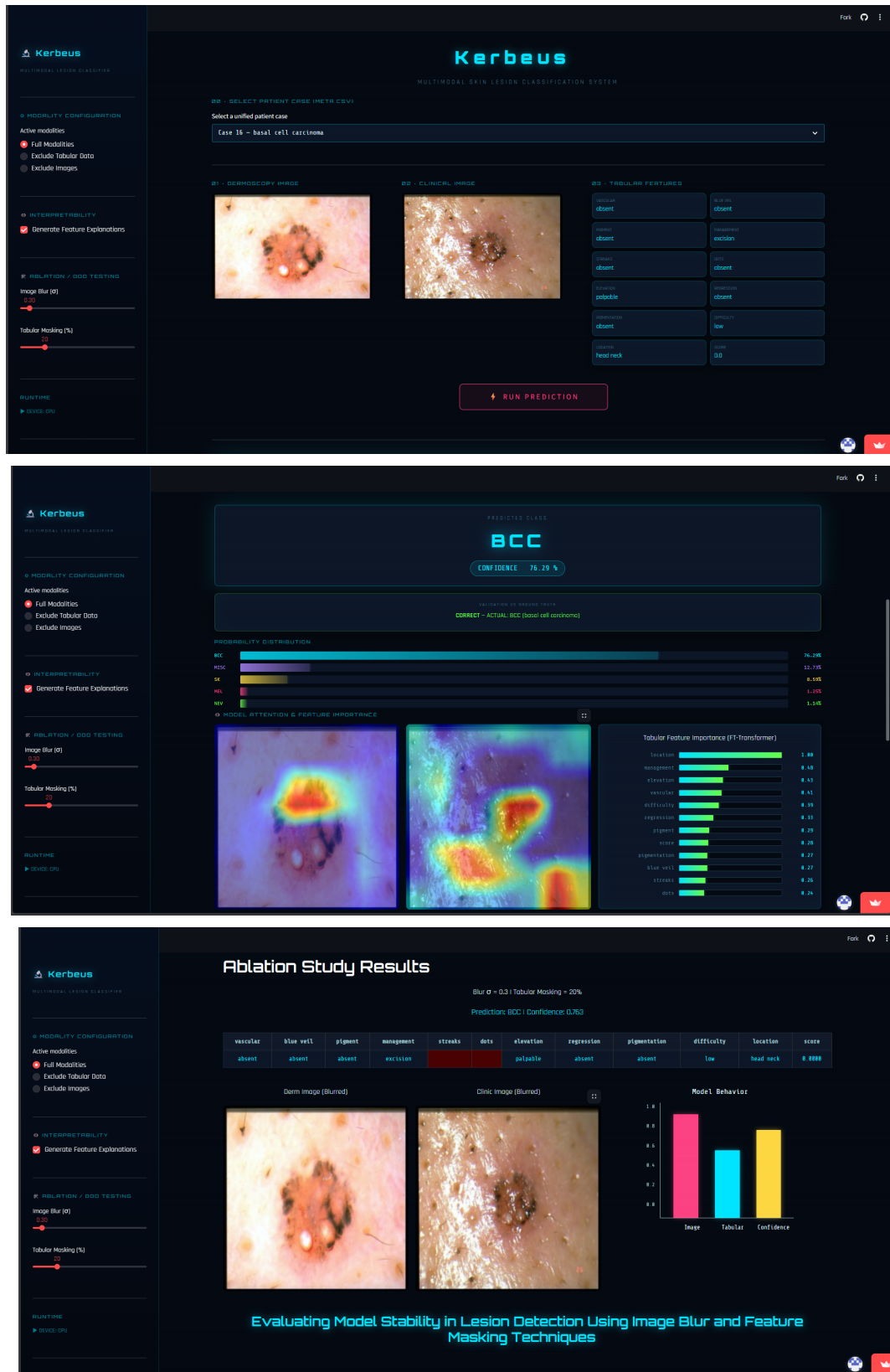


Fig. 9. Dashboard visualizations of Kerbeus : tested on basal cell carcinoma BCC with image blur - 0.30 and tabular masking of 20% image 3 shows the gradient flow from image and metadata .